

AYUSH MAURYA

ayushmaurya2003@gmail.com | ayush.maurya@research.iiit.ac.in | +91-7985149173 | [linkedin.com/in/ayush-maurya](https://www.linkedin.com/in/ayush-maurya) | github.com/Ayush12358 | ayushmaurya.online

ABOUT ME

CS + Computational Human Sciences dual-degree research student at IIIT Hyderabad, working at the intersection of software systems, machine learning, and human behavior - from agent-based electoral simulations to agentic AI platforms. My work spans NLP research, full-stack web apps, product design, and digital humanities.

EDUCATION

IIIT Hyderabad — Dual Degree (B.Tech CSE + M.S. Research) 2021 – Present
Human Sciences Research Centre (HSRC). Supervisor: Dr. Aniket Alam.

- Research: Political Narratives & Electoral Systems in India

Relevant Coursework & Labs 2022 – 2024
Design & Analysis of Software Systems · Human Science Lab (NLP Research)

Mary Gardiner's Convent School (XII: 90.1%, 2021) · City Montessori School (X: 94.2%, 2019)

EXPERIENCE

SERC Digital Twin Environmental IoT
Real-time environmental monitoring dashboard using MERN stack, integrating local IoT sensors for temperature, humidity, water quality, and air quality tracking.
MERN IoT SQL

tafea System Design
Designed a student client software system using SDLC methodologies. Integrated LLM APIs for natural language interfaces.
System Design UI Design LLM

KEY PROJECTS

ElectoralSim [PyPI](#) ↗
High-performance agent-based electoral simulation toolkit. Simulates 1M+ voter agents with 89× speedup via Polars & Numba, supports 8+ electoral systems across 11 country presets, models voter psychology with social network opinion dynamics.
Python Agent-Based Modeling Polars Numba

IPL Strategy Lab ipl.ayushmaurya.online ↗
Agentic AI cricket analytics with multi-agent architecture (Analyst → Strategist → Evaluator) and self-healing code loop. Runs Python/Pandas in-browser via Pyodide (WASM) over 260K+ deliveries.
React Agentic AI Pyodide (WASM) Pandas

CrowdTwin crowdtwin.vercel.app ↗
Predictive crowd monitoring dashboard prototype built for a Tech Product Entrepreneurship course, with idea refinement and pitching to guest evaluators.
Next.js Product Design Prototype

Design for Social Innovation (TAFEA)
AI-assisted platform to help Teach For India fellows plan and customize extracurricular curricula for resource-constrained classrooms.
Human-Centered Design Education AI

TempusLogic [GitHub](#) ↗
NLP diagnostic reasoning suite testing LLM logical consistency under adversarial prompting and contextual shifts via Dyads/Triads reasoning and GSM8K math stress tests.
Python NLP Research LLM Evaluation

My Research — Representing India [research](#) ↗
Computational human sciences research on political narratives & electoral dynamics in India, conducted at IIIT-H's HSRC under Dr. Aniket Alam.
Computational Social Science Political Narratives Research

EXPERTISE

Languages
Python, C/C++, JavaScript/TS, SQL

ML & Data
PyTorch, Agentic AI, Polars, Numba, Pandas, Scikit-learn

Full Stack
React, Next.js, Flutter, Tailwind CSS, PWA, Pyodide (WASM), Bun

Systems
Linux, System Design, Systems Programming

FOR FUN

Bass Guitar & Ukulele
Bass guitarist in college band, ukulele player

Sports
Lawn tennis & pickleball player

OTHER PROJECTS

ML for Local Geoguesser
ResNet50-based location estimation model. Predicts region, lat/lon, and camera angle — 92% region accuracy.
Python PyTorch ResNet50

Age Prediction Model
Transfer learning vs. ensemble methods (Random Forests, SVMs) for age prediction from images.
Machine Learning PyTorch

Highland History Project
OCR & LLM-assisted digitization of Himalayan trade registers. Collaboration with Dr. Aniket Alam at HSRC.
Machine Learning OCR LLM

Campus Mart
UI/UX for a buy-sell-rent campus app with NFC-based exchange and tracking.
UI/UX Design Prototyping

Katha Marketplace
Storytelling-led marketplace prototype for local artisans with a stories-first visual interface.
UI/UX Design Wireframing

WorDrop ↗
Privacy-first offline Android safety app using Vosk on-device speech recognition with configurable hotword triggers.
Flutter Vosk Android

amReader ↗
Client-side M4B audiobook player with chapter parsing, subtitle autoscroll, and IndexedDB offline storage.
React 19 IndexedDB PWA

WebTTS ↗
Browser-based EPUB-to-audiobook converter with dual TTS engines (Web Speech API + Google Cloud).
React Web Speech API PWA